Session 06: Limiting Reagent & Percent Yield

> **Topic:** When two reactants compete, which one runs out — and how much product can you really make?
> **Time:** 150 minutes
> **Big Fat Notebook:** Chapter 7

Opening Attack

Try this:

> You react **25.0 g of N_2 (nitrogen gas)** with **5.00 g of H_2 (hydrogen gas)** to make ammonia:
>
> $$N_2 + 3H_2 \rightarrow 2NH_3$$
>
> (a) Which reactant runs out first?
>
> (b) What is the **maximum** mass of NH_3 (ammonia) you can produce?
>
> (c) If you actually collect **22.1 g** of NH_3, what is the percent yield?

Stuck? Good. By the end of this session, you'll identify the limiting reagent in your sleep, calculate theoretical yield, and judge how efficient your reaction really was.

Worked Pattern

Example 1 — Identifying the Limiting Reagent (Method 1: mol-to-mol comparison)

Problem: $N_2 + 3H_2 \rightarrow 2NH_3$. You have 25.0 g N_2 and 5.00 g H_2. Which is limiting?

Step 1: Convert both given masses to moles.

> Molar mass $N_2 = 2 \times 14.01 = 28.02 \text{ g/mol}$
>
> $\text{mol } N_2 = 25.0 \text{ g} \div 28.02 \text{ g/mol} = 0.8922 \text{ mol } N_2$
>
> Molar mass $H_2 = 2 \times 1.008 = 2.016 \text{ g/mol}$
>
> $\text{mol } H_2 = 5.00 \text{ g} \div 2.016 \text{ g/mol} = 2.4802 \text{ mol } H_2$

Step 2: Pick one reactant and ask: "How much of the other reactant does this need?"

> Using N_2 as reference: 0.8922 mol N_2 needs $0.8922 \times 3 = 2.6766 \text{ mol } H_2$.
>
> We **have** 2.4802 mol H_2. We **need** 2.6766 mol H_2.
>
> $2.4802 < 2.6766 \ \rightarrow$ Not enough H_2! H_2 is **limiting**.

Step 3: Verify from the other direction.

> Using H_2 as reference: 2.4802 mol H_2 needs $2.4802 \div 3 = 0.8267 \text{ mol } N_2$.
>
> We **have** 0.8922 mol N_2. We **need** 0.8267 mol N_2.
>
> $0.8922 > 0.8267 \ \rightarrow$ More than enough N_2. N_2 is **excess**.

Answer: $\boxed{H_2 \text{ is the limiting reagent. } N_2 \text{ is in excess.}}$

Example 2 — Limiting Reagent: Method 2 (Product-Based)

Problem: Same reaction. 25.0 g N_2, 5.00 g H_2. Find the LR by calculating how much NH_3 each reactant *could* produce.

Step 1: mol of each (same as Example 1).

$$\text{mol } N_2 = 0.8922 \text{ mol, mol } H_2 = 2.4802 \text{ mol}$$

Step 2: Calculate mol NH_3 each reactant *could* make.

From N_2: $0.8922 \text{ mol } N_2 \times \frac{2 \text{ mol } NH_3}{1 \text{ mol } N_2} = 1.7844 \text{ mol } NH_3$

From H_2: $2.4802 \text{ mol } H_2 \times \frac{2 \text{ mol } NH_3}{3 \text{ mol } H_2} = 1.6535 \text{ mol } NH_3$

Step 3: The reactant that makes **less** product is limiting.

$1.6535 < 1.7844 \ \rightarrow\ H_2$ makes less NH_3. **H_2 is limiting.**

Answer: $\boxed{H_2 \text{ is limiting (makes only } 1.6535 \text{ mol } NH_3).}$

> **Both methods agree.** Method 1 compares moles; Method 2 compares product. Use whichever feels more natural.

Example 3 — Theoretical Yield from the Limiting Reagent

Problem: Using the LR from Example 1, what is the maximum mass of NH_3 you can produce?

Step 1: Start with moles of the **limiting** reagent.

$\text{mol } H_2 = 2.4802 \text{ mol}$ (limiting)

Step 2: Convert to moles of product using mole ratio.

$$\text{mol } NH_3 = 2.4802 \text{ mol } H_2 \times \frac{2 \text{ mol } NH_3}{3 \text{ mol } H_2} = 1.6535 \text{ mol } NH_3$$

Step 3: Convert moles of product to grams.

Molar mass $NH_3 = 14.01 + 3(1.008) = 17.034 \text{ g/mol}$

$\text{g } NH_3 = 1.6535 \text{ mol} \times 17.034 \text{ g/mol} = 28.17 \text{ g}$

Sig figs: 5.00 (3 sig figs) and 25.0 (3 sig figs) $\rightarrow$ 3 sig figs.

Answer: $\boxed{28.2 \text{ g } NH_3 \text{ (theoretical yield)}}$.

Example 4 — Percent Yield

Problem: The reaction actually produces 22.1 g of NH_3. What is the percent yield?

Step 1: Recall theoretical yield from Example 3.

Theoretical yield = $28.17 \text{ g } NH_3$

Step 2: Divide actual by theoretical.

$$\begin{aligned} \text{Percent yield} &= \frac{\text{actual}}{\text{theoretical}} \times 100\% \\ &= \frac{22.1 \text{ g}}{28.17 \text{ g}} \times 100\% \end{aligned}$$

Step 3: Calculate.

$22.1 \div 28.17 = 0.7845$

$0.7845 \times 100\% = 78.45\%$

Sig figs: 22.1 (3 sig figs) $\rightarrow 78.5\%$

Answer: $\boxed{78.5\% \text{ yield}}$.

Example 5 — Excess Reagent Remaining

Problem: How many grams of N_2 are left over after the reaction?

Step 1: How much N_2 was actually consumed?

> From the LR (H_2): $2.4802 \text{ mol } H_2 \times \frac{1 \text{ mol } N_2}{3 \text{ mol } H_2} = 0.8267 \text{ mol } N_2 \text{ consumed}$

Step 2: Subtract consumed from available.

> Available: $0.8922 \text{ mol } N_2$
>
> Consumed: $0.8267 \text{ mol } N_2$
>
> Remaining: $0.8922 - 0.8267 = 0.0655 \text{ mol } N_2$

Step 3: Convert remaining moles to grams.

> $\text{g } N_2 \text{ remaining} = 0.0655 \text{ mol} \times 28.02 \text{ g/mol} = 1.835 \text{ g}$

Answer: $\boxed{1.84 \text{ g } N_2 \text{ left unreacted}}$.

⚠️ THE TRAP — Picking the Reactant with Fewer Grams

The wrong way:

> 25.0 g N_2 vs. 5.00 g H_2. "H_2 has fewer grams, so H_2 must be limiting." ← This happens to be correct here, but the reasoning is **WRONG!**

Why is this wrong? Grams don't determine the limiting reagent — **moles relative to the**

coefficient do. Consider a counterexample:

> $2H_2 + O_2 \rightarrow 2H_2O$. 10.0 g H_2 vs. 10.0 g O_2.
>
> By grams: equal mass → "neither is limiting"? Wrong!
>
> $\text{mol } H_2 = 10.0 \div 2.016 = 4.96 \text{ mol}$. Needs $4.96 \div 2 = 2.48 \text{ mol } O_2$.
>
> $\text{mol } O_2 = 10.0 \div 32.00 = 0.3125 \text{ mol}$. We have only 0.3125 — far less than the 2.48 needed.
>
> O_2 **is limiting**, even though both started with the same mass!

The right way: Always compare **moles available ÷ coefficient**. The reactant with the smaller ratio is limiting.

> $$\frac{\text{mol } N_2}{\text{coeff } N_2} = \frac{0.8922}{1} = 0.8922$$
> $$\frac{\text{mol } H_2}{\text{coeff } H_2} = \frac{2.4802}{3} = 0.8267$$
> $0.8267 < 0.8922 \quad \rightarrow \quad H_2$ is limiting. ✓

Example 6 — Limiting Reagent: When It's Close

Problem: $4Fe + 3O_2 \rightarrow 2Fe_2O_3$. You have 50.0 g Fe (iron) and 25.0 g O_2 (oxygen gas). Which is limiting?

Step 1: Convert to moles.

> $\text{mol Fe} = 50.0 \div 55.85 = 0.8953 \text{ mol Fe}$
>
> $\text{mol } O_2 = 25.0 \div 32.00 = 0.78125 \text{ mol } O_2$

Step 2: Divide by coefficients.

> Fe ratio: $0.8953 \div 4 = 0.2238$

O_2 ratio: $0.78125 \div 3 = 0.2604$

$0.2238 < 0.2604 \;\rightarrow$ **Fe is limiting.**

Step 3: Verify by product method.

From Fe: $0.8953 \times \frac{2}{4} = 0.44765 \text{ mol } Fe_2O_3$

From O_2: $0.78125 \times \frac{2}{3} = 0.52083 \text{ mol } Fe_2O_3$

Fe makes less. ✓

Answer: $\boxed{Fe \text{ is limiting. Theoretical yield } = 0.44765 \text{ mol } Fe_2O_3 = 71.5 \text{ g } Fe_2O_3}$.

Example 7 — Combined: Limiting Reagent + Percent Yield

Problem: $2Al + 3Cl_2 \rightarrow 2AlCl_3$. 10.0 g Al reacts with 20.0 g Cl_2. 28.5 g $AlCl_3$ is collected. Find the percent yield.

Step 1: Find the limiting reagent.

$\text{mol Al} = 10.0 \div 26.98 = 0.37064 \text{ mol}$

$\text{mol } Cl_2 = 20.0 \div 70.90 = 0.28209 \text{ mol}$

Divide by coefficients: Al = $0.37064/2 = 0.1853$, $Cl_2 = 0.28209/3 = 0.09403$

$0.09403 < 0.1853 \;\rightarrow$ Cl_2 **is limiting.**

Step 2: Theoretical yield from Cl_2.

$\text{mol } AlCl_3 = 0.28209 \times \frac{2}{3} = 0.18806 \text{ mol}$

> Molar mass $AlCl_3 = 26.98 + 3(35.45) = 133.33 \text{ g/mol}$
>
> $\text{g } AlCl_3 = 0.18806 \times 133.33 = 25.07 \text{ g}$ (theoretical)

Step 3: Percent yield.

> $\% \text{ yield} = \frac{28.5}{25.07} \times 100\% = 113.7\%$

Wait — **113.7%?!** That's over 100%. This means either the product is impure (contains unreacted starting material or solvent), or there's a measurement error. Percent yields above 100% are a red flag in real lab work.

Answer: $\boxed{113.7\% \text{ yield — suspiciously high; product likely impure.}}$

Example 8 — The Full Exam Problem

Problem: $CaCO_3 + 2HCl \rightarrow CaCl_2 + CO_2 + H_2O$. 15.0 g $CaCO_3$ (limestone) reacts with 100.0 mL of 2.00 M HCl (hydrochloric acid). (a) Which is limiting? (b) Theoretical yield of CO_2 in grams? (c) If 5.80 g CO_2 is collected, what is the percent yield? (d) How many grams of excess reagent remain?

(a) Limiting reagent:

> $\text{mol } CaCO_3 = 15.0 \div 100.09 = 0.14987 \text{ mol}$
>
> $\text{mol HCl} = 0.1000 \text{ L} \times 2.00 \text{ mol/L} = 0.2000 \text{ mol}$
>
> Divide by coefficients: $CaCO_3 = 0.14987/1 = 0.14987$, HCl = $0.2000/2 = 0.1000$
>
> $0.1000 < 0.14987 \rightarrow$ **HCl is limiting.**

(b) Theoretical yield of CO_2:

$$\text{mol } CO_2 = 0.2000 \text{ mol HCl} \times \frac{1 \text{ mol } CO_2}{2 \text{ mol HCl}} = 0.1000 \text{ mol } CO_2$$

$$\text{g } CO_2 = 0.1000 \times 44.01 = 4.401 \text{ g}$$

(c) Percent yield:

$\%\text{ yield} = \frac{5.80}{4.401} \times 100\% = 131.8\%$ — again suspicious, but that's the calculation.

(d) Excess $CaCO_3$ remaining:

$CaCO_3$ consumed = $0.2000 \times \frac{1}{2} = 0.1000 \text{ mol}$

$CaCO_3$ remaining = $0.14987 - 0.1000 = 0.04987 \text{ mol}$

$$\text{g remaining} = 0.04987 \times 100.09 = 4.99 \text{ g}$$

Answer:

$\boxed{\text{(a) HCl limiting} \quad \text{(b) 4.40 g } CO_2 \quad \text{(c) } 132\% \quad \text{(d) 4.99 g } CaCO_3 \text{ remaining}}$.

What We Just Did — The Limiting Reagent Procedure

Step 1: Convert ALL given quantities to moles.

Grams? Divide by molar mass.

Solution volume + molarity? $\text{mol} = M \times V(\text{L})$.

You cannot compare grams directly. Everything must be in moles.

Step 2: Divide each mole value by its coefficient. The smallest ratio = limiting.

> $$\text{Ratio}_A = \frac{\text{mol}_A}{\text{coeff}_A} \qquad \text{Ratio}_B = \frac{\text{mol}_B}{\text{coeff}_B}$$
>
> The reactant with the **smaller** ratio is the limiting reagent.
>
> Alternative (Method 2): Calculate how much product each reactant *could* make. The reactant that makes **less** product is limiting.

Step 3: Use the LIMITING reagent for ALL further calculations.

> Theoretical yield: start from moles of LR → moles of product → grams of product.
>
> Percent yield: $\frac{\text{actual}}{\text{theoretical}} \times 100\%$.
>
> Excess remaining: calculate how much of the excess reagent was consumed by the LR, then subtract from the original amount.

Copy this. Never guess the limiting reagent by grams again.

💡 Pro Tips — Calculator Shortcuts & Mental Checks

Tip 1 — The fastest LR method. Divide $\text{mol} \div \text{coefficient}$ for each reactant. The **smallest** number wins. This takes 10 seconds and works for any number of reactants.

> $0.8922/1 = 0.892$ vs. $2.480/3 = 0.827 \rightarrow 0.827 < 0.892 \rightarrow H_2$ is LR. Done.

Tip 2 — Percent yield > 100% is a red flag. Unless you made a math error or your product is wet/impure, yields don't exceed 100%. On exams, if you get >100%, double-check your limiting reagent — you probably used the wrong one.

Tip 3 — Excess reagent shortcut. Instead of calculating moles remaining then converting to grams: $\text{g remaining} = \text{g initial} - \text{g consumed}$. Calculate grams consumed from the LR via stoichiometry.

Tip 4 — "Equal masses" trap. When two reactants have the same mass, the one with the **larger molar mass** tends to be limiting (fewer moles), but you MUST still divide by the coefficient. Same mass of O_2 (32 g/mol) vs. H_2 (2 g/mol): O_2 has 16× fewer moles — it's almost certainly limiting.

Tip 5 — Theoretical yield sanity check. Your theoretical yield must be less than the sum of both reactants' masses (conservation of mass) AND must come only from the LR. If your "theoretical yield" exceeds the mass of either reactant alone (for a non-combustion reaction), you used the wrong reagent.

Basic Drills (B1–B10)

> Goal: Automate LR identification and theoretical yield.

B1. $N_2 + 3H_2 \rightarrow 2NH_3$. 14.0 g N_2 + 3.00 g H_2. (a) Find LR. (b) Theoretical yield of NH_3 in grams. (N = 14.01, H = 1.008)

B2. $2H_2 + O_2 \rightarrow 2H_2O$. 8.00 g H_2 + 48.0 g O_2. (a) Find LR. (b) Grams of H_2O produced. (H = 1.008, O = 16.00)

B3. $4Fe + 3O_2 \rightarrow 2Fe_2O_3$. 100.0 g Fe + 50.0 g O_2. (a) Find LR. (b) Grams of Fe_2O_3 produced. (Fe = 55.85, O = 16.00)

B4. $2Al + 3Cl_2 \rightarrow 2AlCl_3$. 5.00 g Al + 15.0 g Cl_2. Find theoretical yield of $AlCl_3$ in grams. (Al = 26.98, Cl = 35.45)

B5. $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$. 10.0 g CH_4 + 30.0 g O_2. (a) Find LR. (b) Grams of CO_2 produced. (C = 12.01, H = 1.008, O = 16.00)

B6. Same reaction as B1. Theoretical yield = 17.0 g. Actual yield = 12.8 g. Find percent yield.

B7. $2Mg + O_2 \rightarrow 2MgO$. 24.3 g Mg + 16.0 g O_2. (a) Find LR. (b) Theoretical yield. (c) If 35.0 g MgO is collected, percent yield? (Mg = 24.31, O = 16.00)

B8. (Excess remaining) Using B3 data: how many grams of the excess reagent remain?

B9. $CaCO_3 \rightarrow CaO + CO_2$. 200.0 g $CaCO_3$ is heated. 100.0 g CaO is collected. Find percent yield. (Ca = 40.08, C = 12.01, O = 16.00)

B10. (Speed round — under 2 min) $2Na + Cl_2 \rightarrow 2NaCl$. 10.0 g Na + 10.0 g Cl_2. (a) LR? (b) Theoretical yield of NaCl in grams. (Na = 22.99, Cl = 35.45)

Advanced Drills (A1–A10)

> Goal: Multi-step integration with Sessions 04–05. Exam traps.

A1. (Multi-step with Session 04) $C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$. 22.0 g C_3H_8 + 80.0 g O_2. (a) Find LR. (b) Theoretical yield of CO_2. (c) How many CO_2 molecules is this? (C = 12.01, H = 1.008, O = 16.00)

A2. (Multi-step with Session 03) A compound is 85.6% C, 14.4% H. Its combustion: $C_xH_y + O_2 \rightarrow CO_2 + H_2O$. (a) Empirical formula? (b) If 10.0 g of this compound burns with 30.0 g O_2, which is limiting? (c) Grams of CO_2 produced?

A3. (Trap-laden) $2H_2 + O_2 \rightarrow 2H_2O$. A student says: "I have 4.00 g H_2 and 32.0 g O_2. 4.00 < 32.0, so H_2 is limiting." What's wrong? Find the actual LR and theoretical yield.

A4. (Reverse engineering) A student reacts 15.0 g N_2 with excess H_2 and collects 16.2 g NH_3. They claim the percent yield is 88.9%. Are they right? Show the calculation and diagnose any error.

A5. (Constructive) Design a limiting reagent problem where the reactants have the SAME mass but one is limiting and the other is in excess. The answer should NOT be obvious from the masses alone. Solve it.

A6. (Constructive) Write a problem where the percent yield is exactly 75.0%. The theoretical yield should be a nice round number. Solve it and verify.

A7. (Table-based) Three reactions with given amounts:

Rxn	Equation	Reactant A	Reactant B
1	$2H_2 + O_2 \rightarrow 2H_2O$	6.00 g H_2	40.0 g O_2
2	$N_2 + 3H_2 \rightarrow 2NH_3$	20.0 g N_2	6.00 g H_2
3	$4Fe + 3O_2 \rightarrow 2Fe_2O_3$	80.0 g Fe	30.0 g O_2

For each: (a) LR, (b) theoretical yield in grams, (c) excess reagent remaining in grams. (H = 1.008, O = 16.00, N = 14.01, Fe = 55.85)

A8. (Timed — under 3 min) $2Al + Fe_2O_3 \rightarrow 2Fe + Al_2O_3$ (thermite). 25.0 g Al + 50.0 g Fe_2O_3. (a) LR? (b) Theoretical yield of Fe? (c) If 30.0 g Fe is collected, percent yield? (d) Excess remaining? (Al = 26.98, Fe = 55.85, O = 16.00)

A9. (Hardest variation) A 5.00 g sample containing both $NaCl$ and $NaBr$ is dissolved and treated with excess $AgNO_3$. All halide ions precipitate as AgCl and AgBr. The total mass of precipitate is 10.38 g. Find the mass percent of NaCl in the original mixture. (NaCl = 58.44, NaBr = 102.89, AgCl = 143.32, AgBr = 187.77 g/mol) *Hint: set up two equations with two unknowns.*

A10. (Exam boss) $2KMnO_4 + 16HCl \rightarrow 2KCl + 2MnCl_2 + 5Cl_2 + 8H_2O$. 25.0 g $KMnO_4$ (potassium permanganate) reacts with 150.0 mL of 3.00 M HCl. (a) Find LR. (b) Theoretical yield of Cl_2 (chlorine gas) in grams AND in liters at STP. (c) If 8.50 L of Cl_2 is collected at STP, percent yield? (d) Grams of excess reagent remaining. (K = 39.10, Mn = 54.94, O = 16.00, H = 1.008, Cl = 35.45)

Now Read

You can now identify the limiting reagent by two methods, calculate theoretical yield, judge percent yield, and find how much excess is left over — at exam speed.

Read **Chapter 7** of *Everything You Need to Ace Chemistry in One Big Fat Notebook*.

You've solved 20 problems, from simple LR identification to reverse-engineering mixture composition from precipitate mass. Limiting reagent problems are now procedural, not puzzling.

> Solutions: solutions/06-solutions.md

📝 **Key Terms — Quick Reference**

Term	What it means
limiting reagent (LR)	the reactant that runs out first — determines theoretical yield
excess reagent	the reactant left over after the reaction completes
theoretical yield	maximum product possible, calculated from the LR
actual yield	what you actually collect in the lab (always ≤ theoretical in a perfect world)
percent yield	$\frac{\text{actual}}{\text{theoretical}} \times 100\%$
mol/coeff method	divide mol of each reactant by its coefficient — **smallest = LR**
product method	calculate product each reactant could make — the one making **less = LR**
grams ≠ LR	never pick LR by comparing grams — always convert to moles first
yield > 100%	red flag — product is impure, wet, or you used the wrong LR
excess remaining	$\text{mol}_{\text{initial}} - \text{mol}_{\text{consumed}}$, then convert to grams